SECTION A

1. (a) An inclined (θ=45°) simple beam is subject to a uniformly distributed load q along AB as shown in Figure 1(a). The total magnitude of vertical load is qL.

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

   (b) A frame is subject to a horizontal force F applied at point B, horizontal force F applied at C and vertical force F applied at D, as illustrated in Figure 1(b). A and E are both pin support and D is a hinge connection (also called moment release).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

2. A simply supported beam with overhang is subject to a vertical point load F_1 on C, a horizontal point load F_2 on C and a uniformly distributed load q along BD as shown in Figure 2(a). The cross-section of the beam is made up of rectangular section, see Figure 2(b).

   (a) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(4 marks)**

   (b) Draw the axial force diagram (AFD) with the sign labels and pertinent values. Note that positive axial force is along the **tensile** direction. **(4 marks)**

   (c) Find the maximum tensile stress, maximum compressive stress and their corresponding position in the beam due to combination of axial force and moment. In question (c), the shear stress induced by shear force is neglected. **(4 marks)**

   (d) Consider shear stress induced by shear force and axial force and moment, draw the stress state at the **centroid** of the left of cross section C. Obtain the **principal** tensile stress and **principal** compressive stress at that location. **(4 marks)**

   (e) Check the strength condition according to the von Mises failure criteria and find the position where the maximum von Mises stress occurs. The allowable von Mises stress $[\sigma_{eff}]$=300 MPa. Check if the beam meets the von Mises failure criteria or not. Check top point, bottom point and center of critical cross sections to ensure safety. Critical sections include both left side of cross section C and right side of cross section C. **(4 marks)**

3. (a) A cantilever shaft is fixed at point A and subject to pure torsion at point B as shown in Figure 3(a). The torque value T=20kN×m, the shaft has hollow cross section with outer

diameter of 200 mm and inner diameter of 190mm as shown in Figure 3(b). The elastic modulus is 200 GPa and Poisson ratio is 0.3. The total length is 2000 mm.

(i) Calculate the shear stress of the shaft at point B. **(5 marks)**

(ii) Calculate the angle of twist at point B (give solution in radians). **(5 marks)**

Hint: because the outer diameter is very close to the inner diameter, it is recommended to assume shear stress is uniform in the shaft.

(b) Consider the truss problem as shown in Figure 3(c). The truss has roller support at A, hinge support at B. All truss members can only sustain axial force. Vertical load of 40 kN and horizontal load of 30 kN are applied at joint C of the truss. The total height of truss is 16m and total width of truss is 12 m. The elastic modulus is 200 GPa and Poisson ratio is 0.3. The cross section of all truss members are shown in Figure 3(d), which is solid circular cross section with diameter of 100 mm.

(i) Calculate the internal force of all truss members. **(4 marks)**

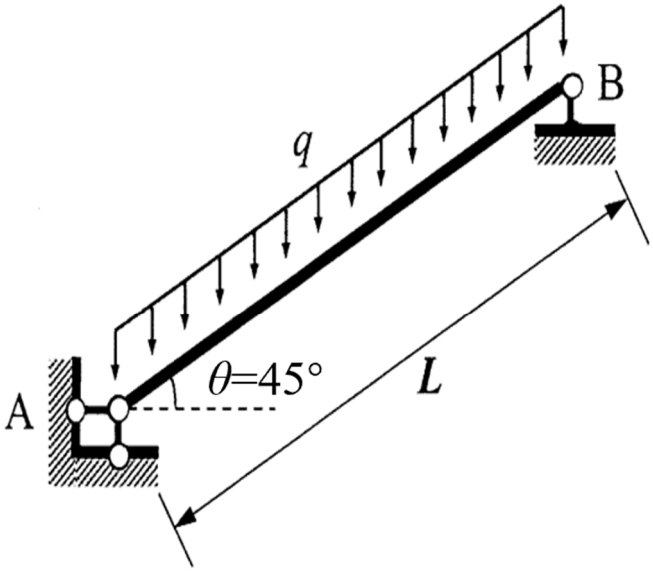
(ii) Check if there are any compression member(s) that will buckle based on Euler's buckling theory. **(6 marks)**

Hint: According to Euler's formula, the critical load for a slender column with span L is

$$P_{cr} = \frac{\pi^2 EI}{(\mu l)^2} \quad \text{(Euler Formula)}$$

The coefficient μ is 1.0 for both pinned ends, μ is 2.0 for one fixed end and one free end, μ is 0.7 for one fixed end and one pinned end, and μ is 0.5 for both fixed ends. In this problem, only consider buckling of single member at a time.

FIGURES



Total vertical load=qL

Figure 1(a). Beam

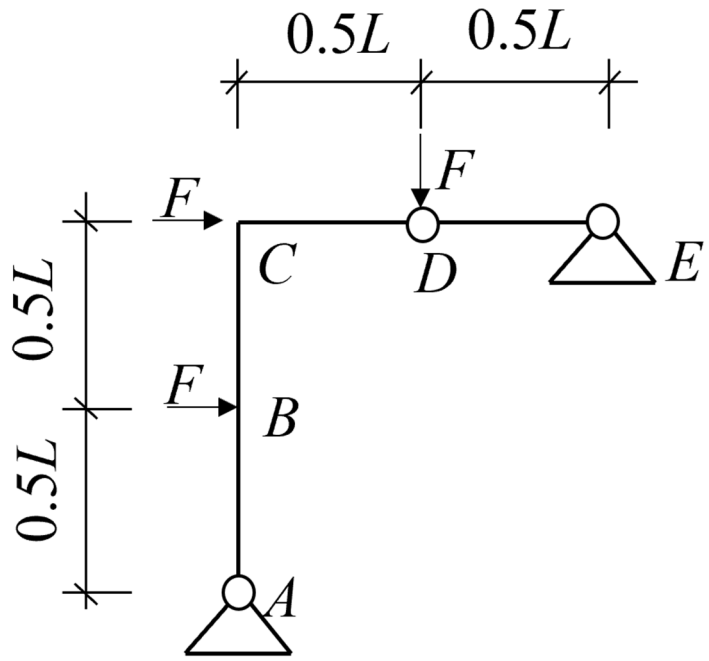


Figure 1(b). Frame

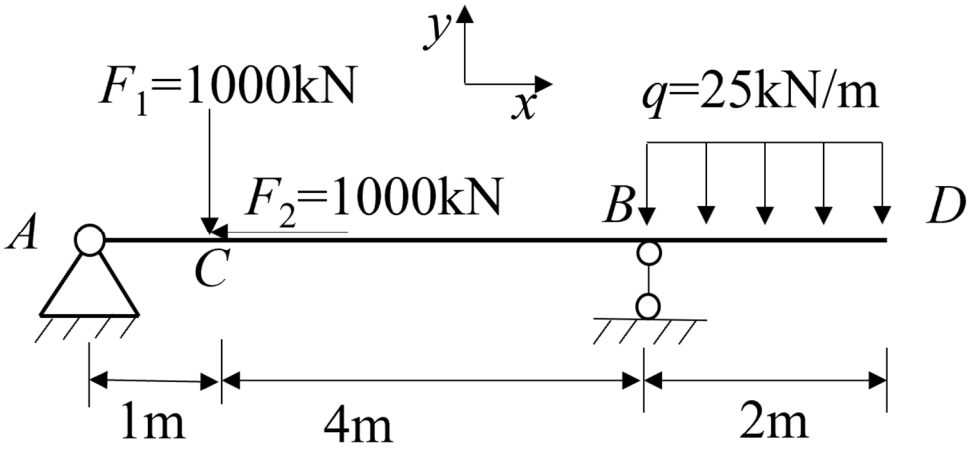


Figure 2(a). Beam with overhang

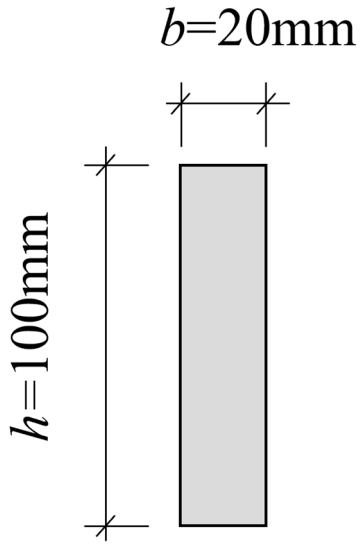


Figure 2(b). Cross section of beam

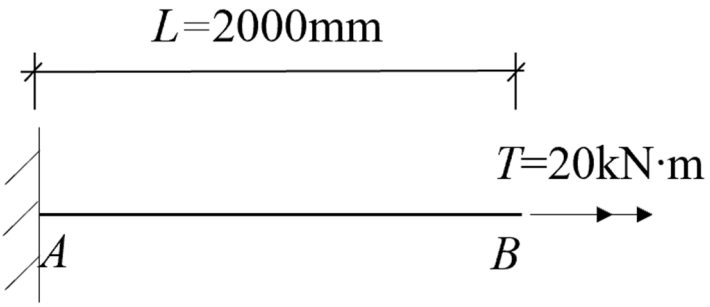


Figure 3(a). Shaft under pure torsion

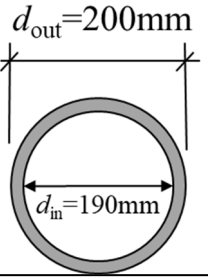


Figure 3(b). Cross section of shaft

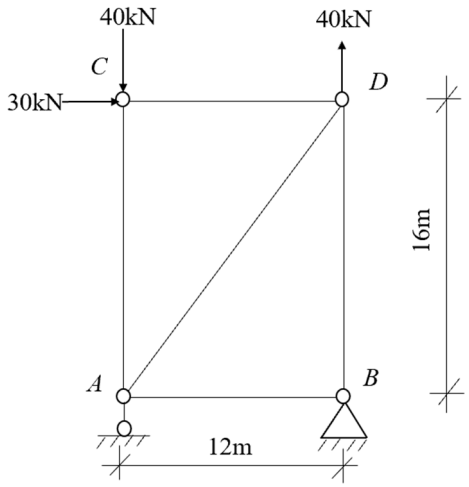


Figure 3(c). Truss

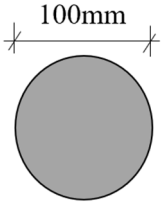


Figure 3(d). Cross section of truss

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